Right Fibrations and Yoneda Lemma

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In this talk we talk about right fibrations, which are a way to understand presheaves into spaces. They will help us understand the Yoneda lemma in the context of higher categories. The first 3 sections are mostly a reminder on the Yoneda lemma in the context of classical category theory. Most of the material can be found in [Ml98]. Section 4 reviews the theory of right fibrations in the context of Segal spaces. Good sources for that are Section 1 of [dB16] and Section 2.1 of [KV14]. The most complete source on left fibations though is probably still Chapter 2 of [Lu09], which discusses everything in the context quasi-categories. The remainder of talk is focused on stating the Yoneda embedding in the context of Segal spaces and for that we strictly follow Sections 2.2 and 2.3 of [KV14].

Classical Yoneda Lemma

Before we actually delve into the main topic, let us review the Yoneda lemma from classical category theory. Let $\mathcal{C}$ be a 1-category and $x \in \mathcal{C}$ an object. We can define a functor, which takes an object y to $Hom_{\mathcal{C}}(y, x)$. It is usually denoted as:

$$\mathcal{Y}_x = Hom_{\mathcal{C}}(-, x) : \mathcal{C}^{op} \to \mathcal{S}\text{et}$$

Note this actually gives us a contravariant functor, as every map $g : y \to z$ gives us a map $g^* : Hom_{\mathcal{C}}(z, x) \to Hom_{\mathcal{C}}(y, x)$ defined by precomposing each map in the domain with g.

Thus for every object x, we have built a contravariant functor $\mathcal{Y}_x$. This construction is actually functorial. Indeed, for any map $f : x \to y$, we have a map $\mathcal{Y}_x \to \mathcal{Y}_y$, which for an object z, simply acts by post-composition $f_* : Hom_{\mathcal{C}}(x, z) \to Hom_{\mathcal{C}}(y, z)$. Thus we get a functor, called the Yoneda embedding functor:

$$\mathcal{Y} : \mathcal{C} \to Fun(\mathcal{C}^{op}, \mathcal{S}\text{et})$$

Now we have following famous lemma:

Lemma 1.1. *(Yoneda Lemma) For a functor $F : \mathcal{C}^{op} \to \mathcal{S}et$ and an object $x \in \mathcal{C}$ an object there is a natural bijection of sets*

$$Nat(\mathcal{Y}_x, F) \xrightarrow{\cong} F(x)$$

By natural we mean the isomorphism is functorial, or in other words, for a map $f : y \to x$, the following diagram commutes

$$\begin{tikzcd} F(y) \arrow[r, "F(f)"] \arrow[d, "\cong"] & F(x) \arrow[d, "\cong"] \\ Nat(\mathcal{Y}_y, F) \arrow[r, "f_*"] & Nat(\mathcal{Y}_x, F) \end{tikzcd}$$

Sketch of Proof. For the proof we realize that there is a specific distinguished element in $\mathcal{Y}_x(x) = Hom_{\mathcal{C}}(x, x)$, namely the famous identity element, id_x. So, for every natural transformation η, the map $\eta_x : \mathcal{Y}_x(x) \to F(x)$, gives us a specific element in $\eta_x(id_x) \in F(x)$. The rest of the proof is showing that η is completely determined by the image of id_x under η_x in $F(x)$. $\square$

Note that for each object $y \in \mathcal{C}$, the functor $\mathcal{Y}_y$ is itself is a contravariant. So, we can apply the result above to get following corollary:

Corollary 1.2. *For two objects $x, y \in \mathcal{C}$ we have following isomorphism:*

$$Nat(\mathcal{Y}_x, \mathcal{Y}_y) \xrightarrow{\cong} Hom_{\mathcal{C}}(x, y)$$

Thus the functor $\mathcal{Y} : \mathcal{C} : Fun(\mathcal{C}^{op}, \mathcal{S}et)$ is fully faithful.

First Efforts in Higher Categories

Now our goal is to generalize this same Yoneda lemma to the realm of higher categories.

Let X be a Segal space. The first difference we note is that for two objects $x, y \in ObjX$, we have a *mapping space* $map_X(x, y)$ and not a mapping set. Thus, if we want to construct our Yoneda embedding, it needs to map our Segal space into presheaves valued in spaces rather than sets.

The naive course of action would be to exactly replicate the steps we discussed above. Concretely, for each object $x \in X$, we define $\mathcal{Y}_x$ to be the map

$$\mathcal{Y}_x : X^{op} \to \mathcal{S}$$

such that it maps $\mathcal{Y}_x(y) = map_X(y, x)$. However, this is exactly where we get stuck. That is because for a map $f : y \to z$, there is *no* map $f^* : map_X(z, x) \to map_X(y, x)$. All that exists is a zig-zag

$$map_X(z, x) \leftarrow map(x, y, z) \xrightarrow{\simeq} map_X(x, y) \times map_X(y, z) \leftarrow map_X(x, y)$$

which would only give us an honest map at the level of homotopy categories. Thus there is simply no way to find an actual map from X^{op} to $\mathcal{S}$ (for more details on mapping spaces in Segal spaces see [Fa17].

Hence we need a different framework to be able to talk about Yoneda lemma.

Second Look at Yoneda Lemma: The Case of Fibrations

Now that we have failed to generalize the Yoneda lemma to case of higher categories, we will try to find a different version of Yoneda lemma in the context of categories that *can* be generalized. For that we need the concept of functors that are *fibered in sets*.

Definition 3.1. A functor $p : \mathcal{D} \to \mathcal{C}$ is fibered in sets if for any map $f : x \to y$ in $\mathcal{C}$ and any object $y' \in \mathcal{D}$ such that $p(y') = y$, there exists a *unique* lift $f' : x' \to y'$ in $\mathcal{D}$, such that $p(f') = f$.

Remark 3.2. If we let $f = id_x \in \mathcal{C}$ and choose a target $x' \in D$ such that $p(x') = x$, then clearly $p(id_{x'}) = id_x$. Thus by uniqueness, the fiber over each x has to be a set.

Example 3.3. The key example for a functor that is fibered in sets is the notion of an over-category. For each object $c \in \mathcal{C}$, we have the over category $\mathcal{C}_{/c}$, which has as objects all maps $d \to c$ (maps with codomain c) and maps commuting triangles

$$\begin{tikzcd} d_1 \arrow[rr] \arrow[dr] & & d_2 \arrow[dl] \\ & c & \end{tikzcd}$$

It is a simple exercise to show that the natural projection map $p : \mathcal{C}_{/c} \to \mathcal{C}$, which takes $d \to c$ to d, is fibered in sets.

But why would we even care about functors that are fibered in sets? The main motivation comes from the *Grothendieck construction* which allows us to construct a functor fibered in sets for each set valued functor.

Definition 3.4. Let $F : \mathcal{C}^{op} \to \mathcal{S}et$ be a functor. We define the category $\int_{\mathcal{C}} F$, called the Grothendieck construction, in the following way

- **Objects**: An object is a tuple (c, x) such that $c \in \mathcal{C}$ is an object and $x \in F(c)$.
- **Morphisms**: For two objects (c, x) and (d, y), we define the maps as

  $$Hom_{\int_{\mathcal{C}} F}((c, x), (y, d)) = \{f \in Hom_{\mathcal{C}}(c, d) : F(f)(y) = x\}$$

  Note here that F is a contravariant functor thus $F(f) : F(d) \to F(c)$.

The category comes with a natural functor $p : \int_{\mathcal{C}} F \to \mathcal{C}$, which maps each tuple (c, x) to c.

Let us see one important example

Example 3.5. Let us take the famous Yoneda functor $\mathcal{Y}_x : \mathcal{C}^{op} \to \mathcal{S}\text{et}$. Applying the Grothendieck construction we get that

$$\int_{\mathcal{C}} \mathcal{Y}_x = C_{/x}$$

Thus, we can in some sense think of $C_{/x}$ as the *representable functor fibered in sets.*

The example suggests following result, which is not too hard to prove:

Theorem 3.6. *For any functor $F : \mathcal{C}^{op} \to \mathcal{S}et$, the Grothendieck construction $p : \int_{\mathcal{C}} F \to \mathcal{C}$ is a functor fibered in sets.*

So, basically we can transfer the whole concept of set valued functors into the language of functors fibered in sets and we even have a notion of a representable functor. With all these tools we can now again state the Yoneda lemma but in this new context.

Lemma 3.7. *Yoneda Lemma for fibered Categories. For any functor fibered in set $p : \mathcal{D} \to \mathcal{C}$, we have following equivalence:*

$$(F_c)^* : Fun_{\mathcal{C}}(\mathcal{C}_{/c}, \mathcal{D}) \xrightarrow{\cong} Fun_{\mathcal{C}}(*, \mathcal{D})$$

where the map comes from precomposing by the functor $F_c : * \to C_{/c}$, which maps the final category (one object and one identity map) to the identity map $id_c : c \to c$.

This approach suggests in order to generalize it to higher categories, we should generalize this concept of fibrations to the realm of higher categories and then try to discuss the Yoneda lemma in the context of higher categories. Hence, we will study *right fibrations.*

Right Fibrations

The idea of a right fibration should generalize the functors fibered in sets we discussed above and should be a way to model contravariant maps into spaces. However, instead of demanding unique lifts we should rather insist of *unique lifts up to homotopy.* So, the goal should be that lifts exists and for each two different lifts, we should be able to lift a homotopy between them. And for each two homotopies between them we must be able to lift a homotopy between those homotopies.

Notation 4.1. Recall that we are using the notation from [Fa17]. In particular, $\Delta[l]$ is the constant simplicial space and $F(n)$ is the discrete simplicial space.

Definition 4.2. A map $f : R \to X$ of simplicial spaces is a right fibration if it is a Reedy fibration and the the following is a homotopy pullback square in the Reedy model structure

$$\begin{tikzcd} R^{F(1)} \arrow[r, "f^{F(1)}"] \arrow[d, "t"'] \arrow[dr, phantom, "\ulcorner"] & X^{F(1)} \arrow[d, "t"] \\ R \arrow[r, "f"'] & X \end{tikzcd}$$

where the map $t : X^{F(1)} \to X$ is the map that takes each arrow to its target. Concretely this means that the map:

$$R^{F(1)} \to X^{F(1)} \times_X R$$

is a Reedy equivalence.

Remark 4.3. Although we stated this definition in the context of Segal space, but the condition is actually quite model independent as mapping objects, arrows and pullbacks can be defined in any model of higher categories.

Remark 4.4. Equivalently, in the context of Segal spaces, we can say a map $f : R \to X$ of simplicial spaces is a right fibration if it is a Reedy fibration and for every $n \in \mathbb{N}$ the following is a homotopy pullback square

$$\begin{tikzcd} R_n \arrow[r, "f_n"] \arrow[d, "{[n]^*}"'] \arrow[dr, phantom, "\ulcorner"] & X_n \arrow[d, "{[n]^*}"] \\ R_0 \arrow[r, "f_0"'] & X_0 \end{tikzcd}$$

where the map $[n]^* : X_n \to X_0$ is the map induced by the map $[n] : [0] \to [n]$ which sends 0 to n.

In other words, for every $n \in \mathbb{N}$, we want following maps

$$R_n \xrightarrow{\simeq} R_0 \times_{X_0} X_n$$

to be trivial Kan fibration.

Let us see some nice examples.

Example 4.5. Let $X = F(0)$ (i.e. the point). Let $R \to F(0)$ be a right fibration over $F(0)$. Then the right fibration condition implies that

$$R_n \xrightarrow{\simeq} R_0 \times_{(F(0))_0} F(0)_n = R_0$$

for every $n \in \mathbb{N}$. So, R is just a *homotopically constant simplicial space* or in other words only has the data of a space. Thus a right fibration over the point is really equivalent to a map from the point into spaces, which is just a space.

Example 4.6. Let us a see a more interesting example. Let $R \to F(1)$ be a right fibration over $F(1)$. We think of $F(1)$ as the discrete simplicial space for which $F(1)_n = Hom([n], [1])$, thus for the first levels we have $F(1)_0 = \{0, 1\}$,

$F(1)_1 = \{00, 01, 11\}$, $F(1)_2 = \{000, 001, 011, 111\}$ ... . Thus we can think of any map into $F(1)$ as a simplicial space of the form:

$$R_{/0} \coprod R_{/1} \Longleftarrow R_{/00} \coprod R_{/01} \coprod R_{/11} \Lleftarrow R_{/000} \coprod R_{/001} \coprod R_{/011} \coprod R_{/111} \Llleftarrow \cdots$$

where we think of $R_{/i}$ as the fiber over i. The right fibration condition implies that any simplicial map

$$R_{/0\ldots1} \to R_{/1}$$
$$R_{/0\ldots0} \to R_{/0}$$

are equivalences. Thus all simplicial levels higher than $F(2)_1$ can be disregarded and the only relevant data is the zig-zag

$$R_{/0} \leftarrow R_{/01} \xrightarrow{\simeq} R_{/1}$$

where the right hand map is again an equivalence by the right fibration condition. Thus the really important part is the map

$$R_{/0} \leftarrow R_{/01}$$

So, we can think of a right fibration over $F(1)$ as a contravariant functor from $[1]$ into spaces, which is exactly what we had hoped for.

Example 4.7. The same example can be expanded to show that every right fibration over $F(n)$ is just the data of a contravariant functor from $[n]$ into spaces.

Now let us look at a very important class of right fibrations.

Example 4.8. Let X be a Segal space. For each object $x \in X$, we get a natural map $i_x : X \to X \times X$, which takes each object y to the tuple (y, x). With that map we can form following pullback square:

$$\begin{tikzcd} X_{/x} \arrow[r] \arrow[d, "p_x"'] \arrow[dr, phantom, "\ulcorner"] & X^{F(1)} \arrow[d, "(s,t)"] \\ X \arrow[r, "i_x"'] & X \times X \end{tikzcd}$$

The simplicial space $X_{/x}$ is also a Segal space. Note that by definition $(X_{/x})_0 = X_1 \times_{X_0} \Delta[0]$, thus at the zero level we just have all maps in X_1 such that the target is x and so we are justified in calling it the *over Segal space* over x.

We have following fact about this map.

Proposition 4.9. *The pullback map $p_x : X_{/x} \to X$ is a right fibration.*

Note that there is a map $x : F(0) \to X_{/x}$, which maps the point to the identity map id_x. We will call this right fibration the *representable right fibration* represented by x. The following theorem justifies our naming convention:

Lemma 4.10. *(Yoneda Lemma for Right Fibrations) Let $R \to X$ be any right fibration. Then the map induced by $* \to X_{/x}$ over X*

$$Map_X(X_{x/}, R) \xrightarrow{\simeq} Map_X(*, R) = * \times_{X_0} R_0$$

is a trivial fibration of Kan complexes.

There is a opposite version of that, which is called a left fibration. The same way we expect right fibrations to model contravariant functors, left fibrations should model covariant functors. We will only give the definition of a left fibration

Definition 4.11. A map $f : L \to X$ of simplicial spaces is a left fibration if it is a Reedy fibration and the the following is a homotopy pullback square in the Reedy model structure

$$\begin{tikzcd} L^{F(1)} \arrow[r, "f^{F(1)}"] \arrow[d, "s"'] \arrow[dr, phantom, "\ulcorner"] & X^{F(1)} \arrow[d, "s"] \\ L \arrow[r, "f"'] & X \end{tikzcd}$$

where the map $s : X^{F(1)} \to X$ is the map that takes each arrow to its source. Concretely this means that the map:

$$L^{F(1)} \to X^{F(1)} \times_X L$$

is a Reedy equivalence.

Remark 4.12. With this definition, we can again discuss left fibations over $F(n)$, which will models a covariant functors from $[n]$ to spaces. Similarly, for each object x in X we can define the under-Segal space $X_{x/}$, which gives us a left fibration $X_{x/} \to X$. Finally, we again have an equivalence

$$Map_X(X_{/x}, L) \xrightarrow{\simeq} Map_X(*, L) = * \times_{X_0} L_0$$

for any left fibration $L \to X$.

The Yoneda Embedding for Higher Categories

Until now we managed to define representable fibrations and use that to state the Yoneda lemma in the context of higher categories. However, we do not have a Yoneda embedding. One hope would be to define a map

$$X \to RFib(X)$$

(where $RFib(X)$ should be the Segal space of right fibrations over X which we have not defined). Moreover, which sends each object x in X to the over-category $X_{/x}$. However, as always, this turns out to be more difficult than

one might expect. Thus we will not approach the Yoneda embedding from this angle.

We will rather follow the path taken in [KV14], which discusses the Yoneda embedding in the context of Segal spaces We will outline the steps in their argument:

1. We define a very interesting model for the complete Segal space of spaces and denote it by $\mathfrak{S}$. (2.2.3 of [KV14])
2. We define a *universal left fibration* over $\mathfrak{S}$, denoted as $\mathcal{E}$, such that for every Segal space Y the pullback map gives us a bijection of sets:

$$Hom(Y, \mathfrak{S}) \xrightarrow{\cong} LFib(Y)$$

   (this is a very limited version of the straightening construction that will be discussed next time). (Lemma 2.2.4 of [KV14])
3. We define a left fibration $\mathcal{M}(X) \to X^{op} \times X$, where $\mathcal{M}(X)$ is supposed to represent the complete Segal space of maps that lies over $X^{op} \times X$. (2.3.2. of [KV14])
4. Using the universality above we get a map $X^{op} \times X \to \mathcal{S}$ (Theorem 2.3.7 of [KV14])
5. Using adjunctions we get a map $j_X : X \to \mathcal{S}^{X^{op}}$. (Theorem 2.3.7 of [KV14])
6. Prove that this map is an embedding! (Theorem 2.3.9 of [KV14])

In the next sections we will discuss some of those steps in greater detail.

Notation 5.1. Before we start I wanted to remind the reader that although we are using the definitions and constructions from [KV14], we will still adhere to the notation that we have introduced in the previous talks.

Remark 5.2. The whole discussion needs some care when it comes to set-theoretical issues (as we will be talking about the complete Segal space of spaces, inside the the category of simplicial spaces). For this talk we will complete ignore are such set-theoretical considerations, but for more details on how this can be fixed see 2.2.2 of [KV14].

The Universal Left Fibration

In this section we define a model for the complete Segal space of space and a left fibration above it and state a Lemma that indicates the left fibration is universal.

Definition 6.1. (2.2.3 of [KV14]) Let $\mathfrak{S}$ be a the simplicial space defined as:

$$\mathfrak{S}_{nm} = \{L \to F(n) \times \Delta[m] : L \text{ is a left fibration}\}$$

Remark 6.2. Note this definition really models spaces. In particular

$$\mathfrak{S}_{00} = \{L \to F(0) \times \Delta[0] : L \text{ is a left fibration}\} = \{L : L \text{ is a space}\}$$

which follows from Example 4.5. Similarly,

$$\mathfrak{S}_{00} = \{L \to F(1) \times \Delta[0] : L \text{ is a left fibration}\} = \{f : \text{ such that } f : L_{/01} \to L_{/1}\}$$

which follows from Example 4.6.

Thus this construction really behaves the way we expected it to do i.e. the 0 level are objects, the 1 level are morphisms and so on.

We have following fact about $\mathfrak{S}$, that justifies the naming "complete Segal space of spaces"

Theorem 6.3. *(Theorem 2.2.11 of [KV14]) The simplicial space $\mathfrak{S}$ is a complete Segal space.*

We now define a left fibration over $\mathfrak{S}$, which we show to be universal.

Definition 6.4. Note that every element $a \in \mathfrak{S}_{nm}$ corresponds to a left fibration $E_a \to F(n) \times \Delta[m]$. Now,l et $\mathcal{E}$ be the following simplicial space

$$\mathcal{E}_{nm} = \coprod_{a \in S_{nm}} (E_a)_{nm}$$

It comes with an obvious projection map $p : \mathcal{E} \to \mathfrak{S}$.

We have following lemma about this projection map

Lemma 6.5. *(Lemma 2.2.4 of [KV14]) The map p is a left fibration. Indeed it is the "universal left fibration" in the sense that, for every simplicial space K, pulling back p induces an isomorphism of sets:*

$$Hom(K, \mathfrak{S}) \xrightarrow{\cong} LFib_{/K}$$

Remark 6.6. As we already pointed out this is a simplified statement of the "straightening" argument as it is relating left fibrations and presheaves valued in statement. The full argument should create an equivalence at the level of homotopy theory rather than just sets.

The Complete Segal Space of Maps

In this section, for every complete Segal space X, we define a complete Segal space $\mathcal{M}$, which is supposed to be the complete Segal space of maps and comes with a natural left fibration to $X^{op} \times X$. As far as I can tell, the construction is not very intuitive and thus it might not be immediatelly clear why it is the right object to consider.

Before we can start we need some combinatorics

Definition 7.1. (2.3.2 of [KV14]) There is a map $\mu : \Delta \to \Delta$ which sends $[n]$ to $[2n+1]$. (For more details on how it behaves on maps see 2.3.2 of [KV14].)

Definition 7.2. For a complete Segal space X, let $\mathcal{M}(X) = \mu^*(X)$. This is often called the *twisted arrow construction*.

Remark 7.3. Unwinding the definitions, we see that the following facts hold:

1. We have
$$\mathcal{M}(X)_n \cong Map(F(n)^{op} \star F(n), X) \cong X_{2n+1}$$
so, it is at each level equivalent to $2n+1$, but in the middle there is a twist in the arrows. (note that the simplicial maps that we omitted in the definition can now just be deduced by using the characterization above).

2. Note that there are obvious maps
$$F(n)^{op} \to F(n)^{op} \star F(n) \text{ and } F(n) \to F(n)^{op} \star F(n)$$
which then induce natural maps:
$$\mathcal{M}(X) \to X^{op} \text{ and } \mathcal{M}(X) \to X$$
which gives us a map
$$\pi_X : \mathcal{M}(X) \to X^{op} \times X$$
This map is a left fibration (Lemma 2.3.4 of [KV14]).

3. Thus, if X is a complete Segal space then $\mathcal{M}(X)$ is also a complete Segal space.

We now have all the building blocks we needed to finally discuss the Yoneda embedding.

The Yoneda Embedding

In this section we will finally state the Yoneda embedding.

Definition 8.1. Let $\mathfrak{P}(X) = \mathfrak{S}^{X^{op}}$ and note it is a CSS.

We can finally state the crucial definition for the Yoneda embedding.

Definition 8.2. (2.3.7. in [KV14]) By Lemma 6.5, $\pi_X : \mathcal{M}(X) \to X^{op} \times X$ corresponds to a map $X^{op} \times X \to \mathfrak{S}$. By adjunction this corresponds to a map
$$j_X : X \to \mathfrak{P}(X)$$
which we call the *Yoneda embedding*.

We have following result which justifies calling it an embedding:

Theorem 8.3. *(Theorem 2.3.9 of [KV14]) The Yoneda embedding $j_X : X \to \mathfrak{P}(X)$ is fully faithful.*

References

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